AKANKSHA TYAGI

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Full stack engineer for AI integrated web apps. Ships across the React/Next.js, FastAPI, and Postgres/pgvector stack with RAG, vector search, and agentic MCP tooling. Master's in computer science, plus published machine learning research.

EDUCATION

UNIVERSITY OF ARKANSAS **GPA: 3.75/4.0** Fayetteville, AR
Master of Science in Computer Science May 2026
Selected Coursework: Image Processing, Privacy Enhancing Technology, Full Stack Deep Learning , Machine Learning.

COLLEGE OF ENGINEERING ROORKEE (COER) Roorkee, India
Bachelor of Technology in Information Technology May 2017

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C/C++
Web: React, Next.js, Node.js, FastAPI, REST APIs, HTML/CSS
AI & ML: LLM integration, RAG, AI agents (MCP), PyTorch, Reinforcement Learning, Graph Neural Networks
Databases: PostgreSQL, MongoDB, Supabase (pgvector)
Cloud & DevOps: Docker, AWS, Git, Linux

PROJECTS

AdvisorDesk — AI-Powered Advisory Content Platform
React · Next.js · TypeScript · FastAPI · Supabase (Postgres + pgvector) · OpenAI · MCP · Docker · AWS
A content hub for financial advisory firms: advisors publish guidance through a CMS, and their clients get instant, cited answers from an AI assistant grounded in that published content.

- Built the full stack: two Next.js frontends (an internal CMS for creating, editing, tagging, and publishing advisory content, plus a client app) on a shared FastAPI backend with Google OAuth, containerized with Docker and deployed on AWS.
- Implemented a RAG assistant that answers client questions over published content using vector search (Supabase pgvector) and OpenAI, returning grounded answers with source citations.
- Engineered an agentic content layer that exposes CMS operations as MCP tools, so a content manager can draft, tag, search, and publish through natural language instead of manual UI steps.

Cybersecurity Lab (UArk Affiliated) Jan 2024 – June 2026
Research Assistantship **Advisor:** Dr. Qinghua Li

- **Reinforcement Learning for Fuzzing:** Developed RL-based models to generate test cases, significantly improving vulnerability detection and code coverage over traditional fuzzing techniques.
- **Master's Thesis** – AI-Based Anomaly Detection in Cyber-Physical Water Testbeds: Built machine learning models for real-time anomaly detection in water systems, enhancing security and resilience of cyber-physical infrastructure.
- **Generative Vision — Diffusion & LDMs:** Built DDPM and latent-diffusion models for high-fidelity image synthesis and text-to-image generation (cross-attention U-Net), evaluated on ALOT, CelebA-HQ, and LAION with FID/IS.

PROFESSIONAL EXPERIENCE & INTERNSHIPS

Reinforcement Learning for Traffic Optimization (IIIT-H Affiliated) Jun 2022 – Dec 2023
Research Internship **Advisor:** Dr. Praveen Paruchuri

- Optimized lane-level dynamics and vehicle-to-vehicle communication by simulating multi-modal traffic environments using SUMO, improving overall traffic control efficiency.
- Reduced traversal times significantly by evaluating RL strategies on grid-world and real-world-inspired road networks, surpassing human-level performance and contributing to a research publication in the International Conference on Vehicle Technology and Intelligent Transport Systems (VEHITS).

CODEVENTURE TECH LLP Roorkee, India
Software Engineer Jun 2021 - Apr 2022

- Designed and developed a city complaint management system using PHP, WordPress, HTML, CSS, and JavaScript, enhancing civic issue reporting and response times.

PUBLICATIONS

- **Tyagi, A.**, Lowalekar, M. and Paruchuri, P., 2024. Improving Lane Level Dynamics for EV Traversal: A Reinforcement Learning Approach. In VEHITS (pp. 134-143). DOI: 10.5220/0012637200003702
  Honors: Nominee for Best Student Paper Award at the VEHITS-2024 conference.
- Uwibambe, M.L., **Tyagi, A.** and Li, Q., 2025, August. A Reinforcement Learning Approach to Multi-Parametric Input Mutation for Fuzzing. In 2025 IEEE International Conference on Cyber Security and Resilience (CSR) (pp. 174-179). IEEE. DOI: 10.1109/CSR64739.2025.11129986